

## NE 470 Nuclear Reactor Theory I

### Group Project #3

Spring 2026

For this project you will develop a numerical solution to a one-dimensional and multi-region reactor composed of various multiplying and non-multiplying “layers” of arbitrary thicknesses. Using two-group diffusion theory, write a computer program that solves the numerical finite-difference diffusion equation to determine the critical dimension of a multi-layered arrangement of various fuels and moderators.

### **Base Configuration (homogeneous core and one reflector) – All Groups**

To model the base homogeneous reactor configuration, use the two-group properties of a typical PWR in the table below. Make sure you use enough nodes to eliminate numerical error.

TABLE 7-2 Few-Group Diffusion Theory Constants for a Typical PWR Reactor Core

| Group Constant | Two-Group |        | Four-Group |          |         |        |
|----------------|-----------|--------|------------|----------|---------|--------|
|                | 1 of 2    | 2 of 2 | 1 of 4     | 3 of 4   | 3 of 4  | 4 of 4 |
| $\nu\Sigma_f$  | .008476   | .18514 | .009572    | .001193  | .01768  | .18514 |
| $\Sigma_f$     | .003320   | .07537 | .003378    | .0004850 | .006970 | .07527 |
| $\Sigma_a$     | .01207    | .1210  | .004946    | .002840  | .03053  | .1210  |
| D              | 1.2627    | .3543  | 2.1623     | 1.0867   | .6318   | .3543  |
| $\Sigma_R$     | .02619    | .1210  | .08795     | .06124   | .09506  | .1210  |

First, assume a “bare core” (no reflector) and determine the bare core critical dimension. Then, add a reflector layer using the following water reflector properties. Determine the “reflector savings” by calculating how much smaller the core region is when adding a reflector.  
 $D_1=1.13$  cm,  $D_2=0.16$ cm,  $\Sigma_{R1}=0.0494$  cm<sup>-1</sup> =  $\Sigma_{s12}$ ,  $\Sigma_{a1}=0.0004$  cm<sup>-1</sup>,  $\Sigma_{a2}=0.0197$  cm<sup>-1</sup>

Your results should report the critical core and reflector dimensions and plot the fast and thermal flux distribution across the core and reflector. Model this as a full core of width WC surrounded by reflectors of width WR on each side (left and right). Use a zero-flux boundary condition at the far left and right edges as well as continuity conditions at the interface between different materials. Normalize your flux to a thermal power of 300 MWth times your group’s number (i.e., Group 7 = 7 x 300 = 2,100 MWth).

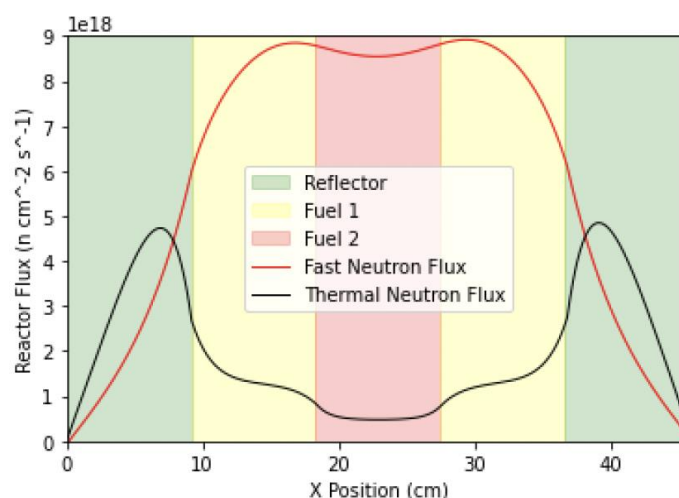
### **Heterogeneous Configuration (multiple regions with variable materials) – Group Specific**

Write your computer program to allow the use any set of properties within each node of the core region (i.e., it should be able to handle different materials in each node and/or group of nodes) so to be able to design a layered core specific to your group. Cross-section sets for each layer option will be provided. Aim to have a minimum of 5 different layers (or more) in your core design. You are encouraged to optimize and modify your design once your program is operational and pursue different arrangements and objectives, as the following suggestions:

1. Flatten flux distribution (reduce peaks)
2. Reduce neutron leakage (reduce fast and/or thermal flux gradient at core’s edge)
3. Maximize flux in interior regions (i.e., for isotope production or irradiation targets)
4. Other considerations you may choose. Your creativity will be rewarded!

The figure below shows an example of a 5-region core with 2 different fuels and a moderator. You will be give cross-sections to use in each region, such as UO<sub>2</sub>, MOX, water, enrichment variations, assemblies with control rods inserted, burned fuel assemblies, beryllium, carbon, etc.

## Report and Presentations due during final exam – Wednesday, May 13, 2026 (3:30-6:00pm)



### Deliverables

- Write a report that includes Introduction, Derivation/Methodology, Results, and Conclusions.
  - Prepare a 10-minute presentation scheduled for the week of final exams. I will provide pizza and soft drinks, so bring your appetite! (please e-mail me for any dietary restrictions)
  - Each member of the group will provide a self-assessment of your own contribution to the group project on a scale from 1 (little or no effort) to 5 (full effort). Sign next to your score and assessment. Include a table with the self-assessment scores in the report.
- **Extra Credit (10%)** – Solve Adjoint Problem and illustrate the validity of the Rayleigh Quotient for small changes in the absorption cross section or solve the problem for 4 energy groups in addition to 2 energy groups. **Graduate students should not use MATLAB and do one of the extra credit problems for regular credit or both to receive extra credit.**

### Group Assignments

| Student                  | NETID     | Group | Student                                                                                                                                                              | NETID     | Group |
|--------------------------|-----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|
| Ocampo, Daniel           | docampo1  | 1     | Graves, Gavin Preston                                                                                                                                                | ggraves8  | 7     |
| Numbers, Jacob Clay      | jnumbers  | 1     | Martinez, Jonathan                                                                                                                                                   | jmart250  | 7     |
| Loo, Matthew             | mloo4     | 1     | Beatty, Conner                                                                                                                                                       | yns374    | 7     |
| Michelson, Trevor Scott  | tmichel2  | 1     | Clevenger, Jacob                                                                                                                                                     | dwp329    | 7     |
| Barrier, Sam             | jbarrie1  | 1     | Haid, Samuel                                                                                                                                                         | shaid     | 7     |
| Deleon, Raven Jade       | rdeleon   | 1     | Dazsi, Luke Joseph                                                                                                                                                   | ldazsi    | 7     |
| Baker, Justin McCain     | jbake118  | 2     | Proctor, Kade                                                                                                                                                        | kprocto6  | 8     |
| Cox, Nathan Charles      | ncox13    | 2     | Noyes, John William                                                                                                                                                  | jnoyes1   | 8     |
| Brady, Elias Walker      | ebrady12  | 2     | Van Deventer, Zacharias Xavier                                                                                                                                       | zvandeve  | 8     |
| Pineda, Luis Daniel      | lpineda   | 2     | Leather, Lauren                                                                                                                                                      | lleathe5  | 8     |
| Lucas, Elizabeth Ann     | elucas5   | 2     | Knueppel, Parker                                                                                                                                                     | pknueppe  | 8     |
| Rashid, Samuel Philip    | srashid6  | 2     | Mason, Elliott Sinclair                                                                                                                                              | emason11  | 8     |
| Clark, Anderson Lee      | aclar157  | 3     | Behrmann, Maxwell Henry                                                                                                                                              | mbehman   | 9     |
| Zondor, Justin Gabriel   | jzondor   | 3     | Wilt, Jude William                                                                                                                                                   | jwilt3    | 9     |
| Gautier, Micah Pierce    | mgautier  | 3     | Riordan, Isaac Kawika                                                                                                                                                | iriordan  | 9     |
| Roberts, Hana Katherine  | hrober52  | 3     | Weber, Michael Callihan                                                                                                                                              | mweber18  | 9     |
| Bolas, Avery Michael     | abolas    | 3     | Maddox, Cole                                                                                                                                                         | cmaddo12  | 9     |
| Gann, Gabriel Thomas     | ggann     | 3     | Hapke, Hunter Aiden                                                                                                                                                  | hhapke    | 9     |
| Liddell, Landon Charles  | lliddet2  | 4     | Roy, Jackson                                                                                                                                                         | jroy7     | 10    |
| Barr, Reagan             | rbarr6    | 4     | Jacob, Max                                                                                                                                                           | mjacob2   | 10    |
| Presutti, Jack Richard   | jpresutt  | 4     | Tate, Myles Denley                                                                                                                                                   | mtate24   | 10    |
| Wachula, Evan            | ewachula  | 4     | De Almeida, Viggo Faria                                                                                                                                              | vdalme1   | 10    |
| Evans, Elijah            | eevens33  | 4     | Gizzie, Hayden Quinn                                                                                                                                                 | hgizzie   | 10    |
| Lorenz, Gregory William  | glorenz   | 4     | German, Michael                                                                                                                                                      | mgerman   | 10    |
| Stapleton, Phoebe Anne   | pstaple2  | 5     | Stewart, Riley                                                                                                                                                       | rstewa46  | 11    |
| Orick, Nathanael         | norick    | 5     | Reilly, Joseph Samuel                                                                                                                                                | jreill10  | 11    |
| Buchanan, Jeremy Levi    | jbucha25  | 5     | Turner, Matthew Grady                                                                                                                                                | mtum100   | 11    |
| Diaz Abreu, Angel        | adiazabr  | 5     | Demella, Max Edward                                                                                                                                                  | mdemella  | 11    |
| Loucks, Greyson Roman    | gloucks   | 5     | Prasanna, Sudarsini                                                                                                                                                  | sprasana1 | 11    |
| Skoba, Dawson Slade      | dskoba    | 5     | Patel, Mansi Sanjay                                                                                                                                                  | mpatel92  | 11    |
| Asquith, London Scott    | lasquith  | 6     | <b>Legend</b><br><b>Graduate Students Group</b><br><b>Team Leader (Submits Assignment)</b><br><b>Team Convener (Communicates Group)</b><br>email: NETID@vols.utk.edu |           |       |
| Serrano, Thomas Berthold | tserran1  | 6     |                                                                                                                                                                      |           |       |
| Dana, Clark Matthew      | cdana     | 6     |                                                                                                                                                                      |           |       |
| Carney, Cameron Gerald   | ccarney11 | 6     |                                                                                                                                                                      |           |       |
| Gudino Meliton, Melissa  | mmeliton  | 6     |                                                                                                                                                                      |           |       |
| Noe, Savannah Grace      | kjo477    | 6     |                                                                                                                                                                      |           |       |